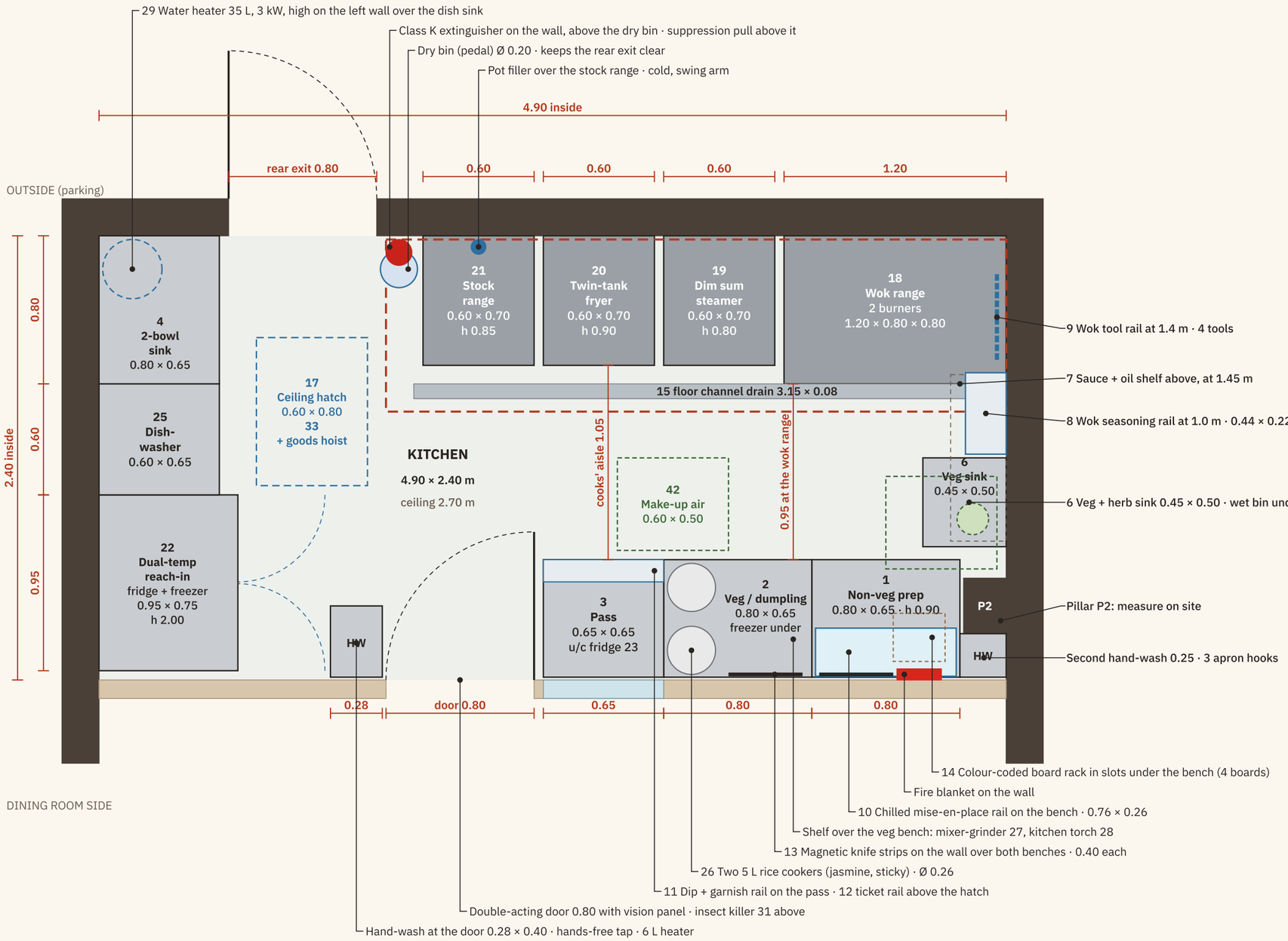
# Kitchen equipment layout

For commercial kitchen equipment vendors · plan at 1:25 on A3 · every item named and sized on the drawing

Cabin East · Wok | Steam | Draught · Concept 27

**Sheet 1 of 2 · A3**

singhgenics.github.io/Simplex/kitchen.html




0 — 1 m — 2 m

**PLAN 1:25 on A3 · dimensions in metres**

Back wall at the top, dining room at the bottom; left and right as you walk in from the road. Dashed red: canopy hood above (3.35 × 0.95, bottom 2.05 m).

Dashed blue: hatch and fridge doors. Dashed green: make-up air diffusers above (2). Dashed grey: shelves above. Heights and services: sheet 2.

## Equipment list: the same numbers 1 to 44 as the BOQ

| NO | EQUIPMENT / TYPE | SIZE W × D × H (mm) | QTY |
|---|---|---|---|
| | STAINLESS STEEL FABRICATION | | |
| 1 | Non-veg prep bench | 800 × 650 × 900 | 1 nos |
| 2 | Dumpling and veg prep bench | 800 × 650 × 900 | 1 nos |
| 3 | Pass counter | 650 × 650 × 900 | 1 nos |
| 4 | Dish sink, 2 bowls | 800 × 650 × 900 | 1 nos |
| 5 | Wall shelf over the dish line | 1,400 × 350 | 2 nos |
| 6 | Veg and herb wash sink | 450 × 500 × 900 | 1 nos |
| 7 | Sauce and oil wall shelf | 900 × 300 | 1 nos |
| 8 | Wok seasoning rail | 440 × 220 | 1 set |
| 9 | Wok tool rail | 470 long | 1 nos |
| 10 | Chilled mise-en-place rail | 760 × 260 × 220 | 1 nos |
| 11 | Dip and garnish rail | 650 × 120 | 1 nos |
| 12 | Ticket rail | 650 long | 1 nos |
| 13 | Magnetic knife strip | 400 long | 2 nos |
| 14 | Cutting-board rack | 280 × 420 × 420 | 1 nos |
| 15 | Floor channel drain | 3,150 × 80 | 3.15 m |
| 16 | Storage deck racking | 3 sizes, see BOQ | 3 nos |
| 17 | Ceiling hatch frame with fold-down ladder | 600 × 800 opening | 1 nos |
| | COOKING, COLD AND KITCHEN EQUIPMENT (BRANDED) | | |
| 18 | Chinese wok range, 2 burners | 1,200 × 800 × 800 | 1 nos |
| 19 | Dim sum steamer, 2 holes | 600 × 700 × 800 | 1 nos |
| 20 | Twin-tank fryer | 600 × 700 × 900 | 1 nos |
| 21 | Stock range, 2 burners | 600 × 700 × 850 | 1 nos |
| 22 | Reach-in, 2 doors, dual temperature | 950 × 750 × 2,000 | 1 nos |
| 23 | Under-counter fridge (pass) | 600 × 550 × 700 | 1 nos |
| 24 | Under-counter freezer, 2 doors | fits under 800 × 650 | 1 nos |
| 25 | Under-counter dishwasher | 600 × 650 × 900 | 1 nos |
| 26 | Rice cooker | Ø 260 | 2 nos |
| 27 | Mixer-grinder | | 1 nos |
| 28 | Kitchen blowtorch | | 1 nos |
| 29 | Storage water heater | Ø 330 × 500 | 1 nos |
| 30 | Water heater for a hand-wash basin | | 2 nos |
| 31 | Insect killer | 500 × 180 | 1 nos |
| 32 | Heat lamp over the pass | | 2 nos |
| 33 | Goods hoist | | 1 nos |
| 34 | Smallware | | 1 lot |
| 35 | Bamboo steamer stack | Ø 260 | 3 nos |
| | HOOD, EXHAUST AND VENTILATION | | |
| 36 | Canopy hood, wall type, with baffle filters | 3,350 × 950 × 550 | 1 nos |
| 37 | Exhaust ducting, rectangular | 400 × 400 | 5 m |
| 38 | Discharge elbow | 400 × 400 | 1 nos |
| 39 | ESP with carbon odour filter | 500 × 1,000 × 900 | 1 nos |
| 40 | Inline exhaust fan | Ø 440 | 1 nos |
| 41 | Fresh-air intake fan | 500 × 480 | 1 nos |
| 42 | Make-up air diffuser, perforated, low velocity | 600 × 500 | 2 nos |
| 43 | Make-up air cooling coil with condenser | condenser 800 × 550 × 300 | 1 set |
| 44 | Exhaust fan and ESP control panel | | 1 nos |
| | Gas, plumbing, electrical, civil, fire: main BOQ | | |

## NOTES

Stainless steel 304 benches, 0.90 m high; hot line 0.70 deep, wok range 0.80. Cooks' aisle 1.05 m (0.95 m at the wok range). Cement-board ceiling at 2.70 m; storage deck above, hoist at the hatch. Pan-Asian menu: about 23 recipes, 40 choices; one chef + one helper. The numbers 1 to 44 on the items are the BOQ numbers (boq.html). Positions are from the 3D model: check walls and pillar P2 on site.

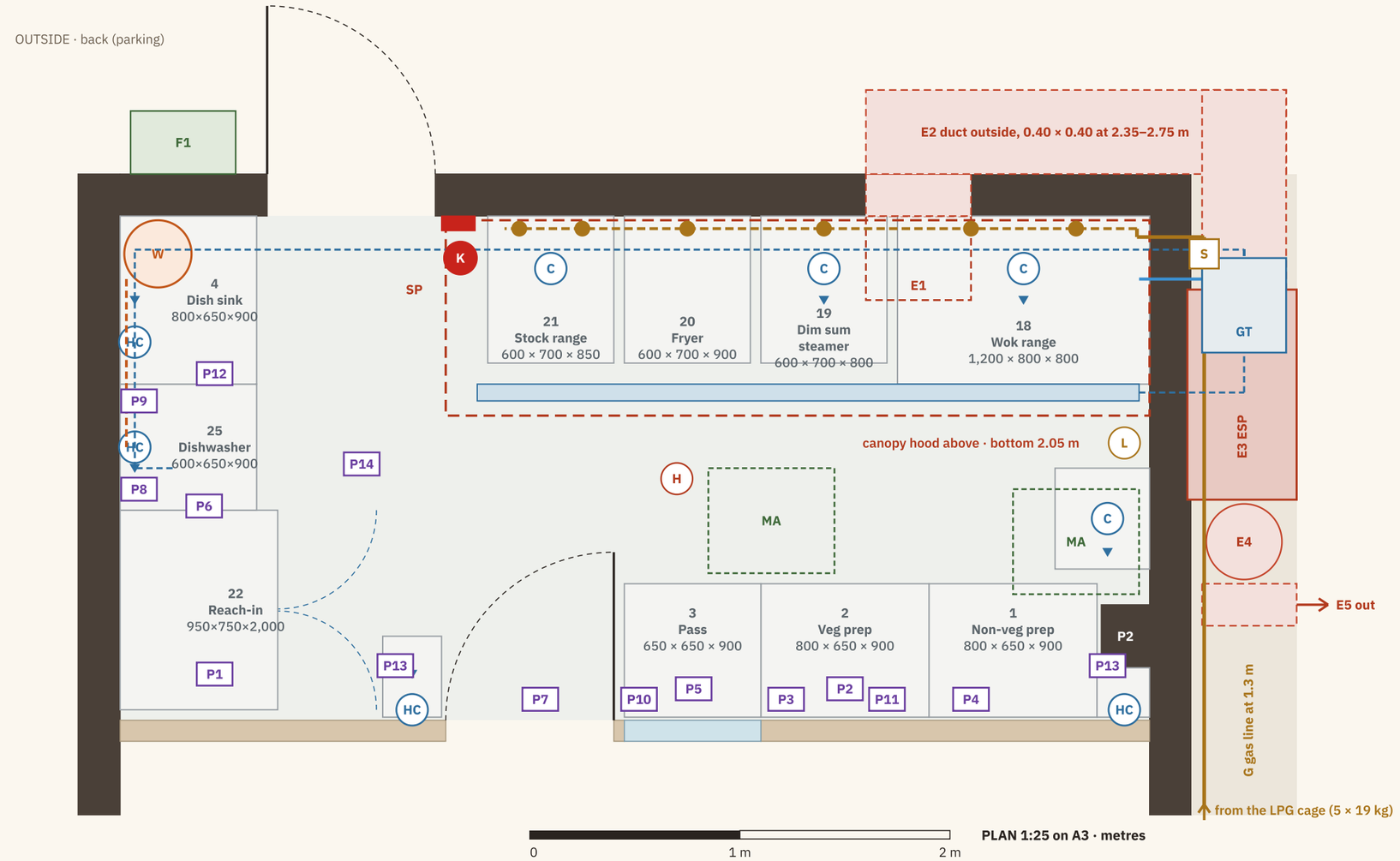
# Kitchen services and exhaust

For kitchen equipment, exhaust, gas, plumbing and electrical vendors · plan at 1:25 and elevation at 1:40 on A3

Cabin East · Wok | Steam | Draught · Concept 27

**Sheet 2 of 2 · A3**

singhgenics.github.io/Simplex/kitchen.html




## Services list

| TAG | ITEM | SIZE / NOTE |
|---|---|---|
| E1 | Duct through the back wall | 0.40 × 0.40 at 2.35–2.75 m |
| E2 | Duct along the wall, round the corner | 0.40 × 0.40, outside |
| E3 | ESP + carbon odour filter | 0.50 × 1.00 × 0.90, 2.05–2.95 m |
| E4 | Inline exhaust fan | Ø 0.44 · on backup |
| E5 | Side discharge, mesh + rain cowl | 0.40 × 0.40 at ~2.6 m |
| F1 | Fresh-air intake fan, back wall | 0.50 × 0.48 at 3.3 m |
| MA | Make-up air diffusers ×2, ceiling | 0.60 × 0.50 each · ~2.4 m/s |
| CC | Cooling coil in the make-up duct | 1.5 TR · 3rd condenser on the wall |
| G | LPG line, copper, from the cage | at 1.3 m, in at the back corner |
| S | Solenoid valve on the entry, outside | interlocked: fan + suppression |
| SP | Hood suppression pull + gas cut-off | at the rear exit, 1.3–1.5 m |
| GT | Grease trap, outside back corner | 0.45 × 0.45 |
| W | Storage water heater, 3 kW | 35 L · left wall over the sink |
| PF | Pot filler, cold, swing arm | over the stock range |
| WC | Water cabinet: pump, filter, softener | outside, 1.2–2.0 m · on backup |
| SU | Water sump, buried | 1,000 L · 1.3 × 1.0 × 0.8 · garden |
| P1 | Reach-in fridge | 1-ph socket · on backup |
| P2 | Under-counter freezer | 1-ph socket · on backup |
| P3 | Two rice cookers | twin 1-ph socket |
| P4 | Chilled mise-en-place rail | 1-ph socket · on backup |
| P5 | Heat lamps over the pass | 1-ph, switched |
| P6 | Dishwasher | 1-ph or 3-ph: confirm model |
| P7 | Insect killer | 1-ph |
| P8 | Hood-light switch | left wall, by panel P9 |
| P9 | Exhaust fan + ESP control panel | left wall over the DW · on backup |
| P10 | Pass under-counter fridge | 1-ph socket · on backup |
| P11 | Mixer-grinder | 1-ph socket, 16 A |
| P12 | Water heater isolator | own 3 kW MCB |
| P13 | Hand-wash heaters ×2 | 1 kW each, 6 L |
| P14 | Goods hoist over the hatch | 200 kg · 0.4 kW |

### EXHAUST + AIR

Allow about 4,000–5,000 m³/h for the whole hood (vendor to size); about 8 m/s in the 0.40 × 0.40 duct. ESP + carbon filter before the fan. No roof stack: apartment above, parking behind. Hood overhang 0.20 m at the open end, 0.15 m over the wok range. Make-up air about 80% of the exhaust through two low-velocity diffusers (~2.4 m/s), cooled by a 1.5 TR coil, interlocked.

### GAS

5 × 19 kg LPG (4 on the manifold + 1 full spare = 95 kg) in a locked cage on the right strip. Typical load: wok 2 × 18, stock 2 × 8, fryer 18, steamer 11 kW = about 80 kW connected, 55 kW at once; 4 cylinders give about 60–75 kW in Goa: vendor to confirm the burner ratings, add a vaporiser if they are higher. A valve per appliance; solenoid shuts off with the fan or the suppression; leak detector low on the right wall.

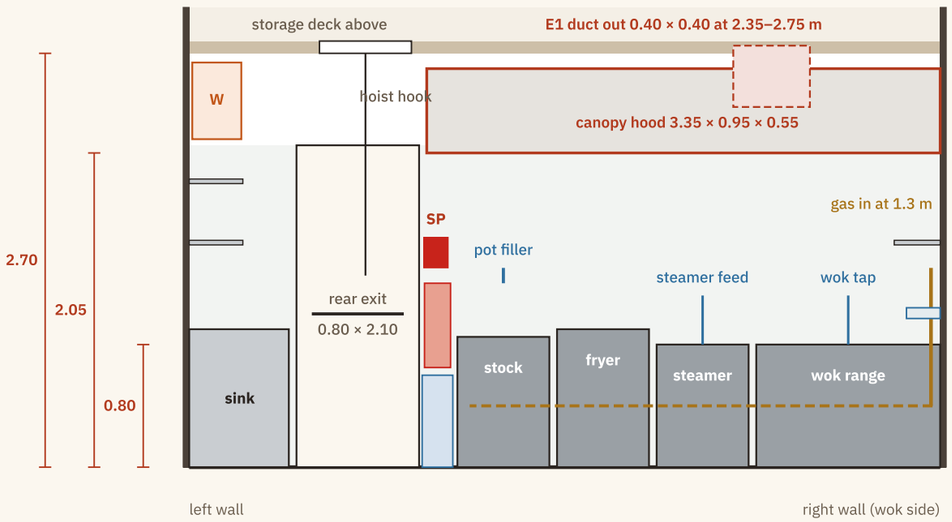
### WATER + DRAINS

Cold: 1,000 L buried sump → pump, filter, softener cabinet (on backup) → kitchen. Hot: 35 L heater for the sink and pre-rinse, 6 L heaters under both hand-wash basins, dishwasher with its own booster. All kitchen drains to the grease trap, then the grey-water line (never the biodigester).

### POWER

Shop supply: three-phase, about 15–20 kW. Backup 7.5 kVA covers the fridges, rails, fan, ESP, lights and the water pump. Not the ACs.

## Hot line elevation · back wall seen from inside · 1:40 on A3




## Legend: every symbol and letter on this sheet

- canopy hood outline (above)
- exhaust duct, outside, 2.35–2.75 m
- gas line, copper
- gas manifold at 0.40 m (dot = valve)
- S solenoid shut-off valve
- L gas leak detector
- HC water point, hot + cold
- C water point, cold only
- drain point
- drain line to the grease trap
- hot water pipe from heater W
- cold water main in (WM)
- GT grease trap (outside, back corner)
- W 35 L storage water heater
- P1 P1 to P14 power point (services list); plain numbers 1 to 44 are BOQ items
- MA make-up air diffusers, ceiling
- H heat detector
- K Class K extinguisher (wok and fryer fires)
- SP hood suppression pull + gas cut-off
- P1, P2 structural pillars on the right wall
- E, F, CC, G, WC, SU, PF: services list